

Deep Learning in Asset Pricing

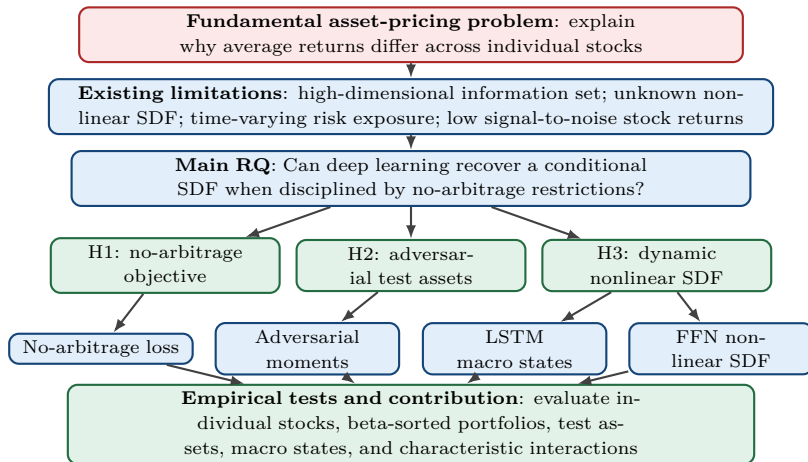
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Framework



Motivation

Why existing approaches struggle

- The SDF may depend on a **large information set**: firm characteristics and macroeconomic variables.
- The functional form is unknown and likely **nonlinear**.
- Risk exposures vary with **firm characteristics, economic conditions, and over time**.
- Individual stock risk premia have a **low signal-to-noise ratio**.

Why return prediction is not enough

Traditional ML models directly predict returns. However, a large amount of volatility in returns comes from unpredictable noise, and directly predicting returns will be affected by low signal-to-noise ratio interference.

Why is this research important

If SDF can be accurately estimated:

1. Explaining the differences in cross-sectional returns of individual stocks.
2. Constructing a mean variance efficient tangent portfolio (SR maximization).
3. Decomposing returns into predictable systematic and unpredictable non-systematic risks.

Research Questions

Main research question

Can deep learning estimate a conditional SDF that explains individual stock returns when disciplined by no-arbitrage restrictions?

1. **Objective:** Does a no-arbitrage learning objective outperform return prediction?
2. **Test assets:** Can adversarially selected test assets improve SDF identification?
3. **Functional form:** Do dynamic macro states and nonlinear characteristic interactions improve asset-pricing performance?

The three sub-questions all derive from the same object: how to recover an SDF from high-dimensional, nonlinear, and time-varying conditioning information.

Contributions

Related literature	What prior studies do	How this paper extends
Machine learning for return prediction	Predict individual stock returns and map predicted returns into asset-pricing implications. Gu et al., (2020); Feng et al., (2018)	Shifts the learning target from prediction errors to no-arbitrage pricing errors .
High-dimensional SDF and factor models	Estimate factor models or SDFs using linear structures, characteristic factors, or no-arbitrage penalties. Fama-French((1993, 2015); Lettau-Pelger(2020)	Builds a nonlinear conditional SDF where weights and loadings depend on firm characteristics and macro states.
GMM and no-arbitrage moments	Use moment restrictions or neural networks with fixed or low-dimensional test assets. Hansen and Jagannathan (1997); Bansal and Viswanathan (1993)	Uses adversarial GMM to select informative moments from an effectively infinite set.
Macroeconomic states	Use macro factors, differenced macro predictors, or low-dimensional factor structures. Hochreiter and Schmidhuber (1997).	Extracts dynamic hidden macro states from 178 macro time series using LSTM.

Hypotheses

H1. No-arbitrage pricing objective

- Models trained on minimizing pricing errors outperforms prediction-based FFN.
- GAN further outperforms linear no-arbitrage benchmarks.

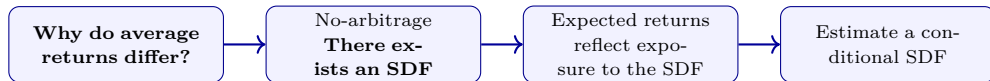
H2. Adversarial test-assets

Adversarially selected characteristic-managed test assets should improve SDF identification and asset-pricing performance.

H3. Dynamic nonlinear SDF

The SDF should perform better when it uses dynamic macroeconomic states and allows nonlinear interactions among firm characteristics.

Why the SDF Matters



No-arbitrage pricing theory

$$E_t[M_{t+1}R_{t+1,i}^e] = 0$$

- If SDF is correct, no pricing error exists.
- Valid models should minimize pricing errors.

Risk-premium implication

$$E_t[R_{t+1,i}^e] = \beta_{t,i}E_t[F_{t+1}] + \epsilon_{t+1,i}$$

- $F_{t+1} = \omega_t^\top R_{t+1}^e$ is traded factor behind SDF.
- $\beta_{t,i}$ measures exposure to systematic risk.

Why this becomes a deep-learning asset-pricing problem

$$M_{t+1} = 1 - \omega_t^\top R_{t+1}^e, \quad \omega_t = E_t[R_{t+1}^e R_{t+1}^{e\top}]^{-1} E_t[R_{t+1}^e]$$

- The SDF weights ω_t and loadings $\beta_{t,i}$ are functions of macro states and firm characteristics.
- Directly estimating the conditional inverse covariance matrix is infeasible for thousands of stocks.
- The paper therefore estimates the SDF indirectly through no-arbitrage moment conditions.

Adversarial GMM: test-asset generator $g(\cdot)$



Conditional moments to test assets

$$E_t[M_{t+1}R_{t+1,i}^e] = 0$$

$$\implies E[M_{t+1}R_{t+1,i}^e g(I_t, I_{t,i})] = 0$$

- Characteristic-managed test assets.

$$\min_{\omega} \max_g \frac{1}{N} \sum_{j=1}^N \left\| E \left[\left(1 - \sum_{i=1}^N \omega(I_t, I_{t,i}) R_{t+1,i}^e \right) R_{t+1,j}^e g(I_t, I_{t,j}) \right] \right\|^2$$

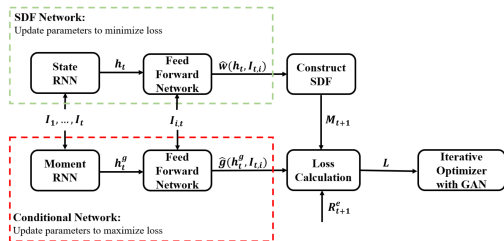
- The **conditioning network** chooses $g(\cdot)$ to maximize pricing errors.
- The **SDF network** chooses ω to minimize those errors.
- Training stops when the adversary can no longer find systematically mispriced test assets.

Why fixed test assets are not enough

- Traditional tests use pre-specified portfolios, such as size–value sorts.
- A model may price those portfolios but still miss other pricing information.

Loss Function and Model Architecture

Model architecture



- The **SDF network** estimates portfolio weights ω .
- The **conditioning network** chooses $g(\cdot)$, the test-asset generator.
- The **LSTM** extracts hidden macroeconomic states from macro time series.

GAN pricing-error loss

$$L(\omega \mid \hat{g}, I_t, I_{t,i}) = \frac{1}{N} \sum_{i=1}^N \frac{T_i}{T} \left\| \frac{1}{T_i} \sum_{t \in T_i} M_{t+1} R_{t+1,i}^e \hat{g}(I_t, I_{t,i}) \right\|^2.$$

- The loss is the average squared pricing error of instrumented stocks.
- $\frac{T_i}{T}$ gives more weight to assets with larger samples.
- For a fixed $\hat{g}$, the SDF network solves: $\hat{\omega} = \arg \min_{\omega} L(\omega \mid \hat{g}, I_t, I_{t,i})$.

Benchmark: return forecasting loss

$$\hat{\mu} = \arg \min_{\mu} \frac{1}{T} \sum_{t=1}^T \frac{1}{N_t} \sum_{i=1}^{N_t} \left(R_{t+1,i}^e - \mu(I_t, I_{t,i}) \right)^2.$$

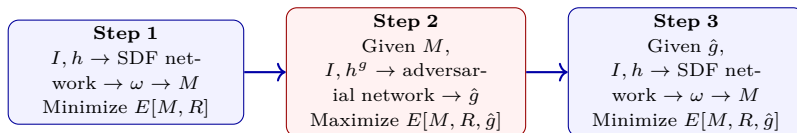
- FFN minimizes prediction errors.
- GAN minimizes no-arbitrage pricing errors.

GAN Training Procedure

The SDF network and the conditioning network play a min-max game:

$$\{\hat{\omega}, \hat{h}_t, \hat{g}, \hat{h}_t^g\} = \min_{\omega, h_t} \max_{g, h_t^g} L(\omega \mid \hat{g}, h_t^g, h_t, I_{t,i}).$$

SDF network reduces pricing errors, **conditioning network searches for test assets that maximize them.**



SDF network

- Inputs: firm characteristics and SDF macro states h_t .
- Output: SDF portfolio weights $\omega(I_t, I_{t,i})$.
- Objective: minimize no-arbitrage pricing errors.

Conditioning network

- Inputs: Adversarial firm characteristics and adversarial macro states h_t^g .
- Output: conditioning instruments $g(I_t, I_{t,i})$.
- Objective: find the hardest-to-price test assets.

Data, Sample, and Benchmark Models

Data and sample

- Universe: all available U.S. stocks from CRSP
- Frequency: monthly excess returns
- Sample period: 1967–2016
- Training: 1967–1986
- Validation: 1987–1991
- Testing: 1992–2016
- Covariates: 46 firm characteristics and 178 macro time series

Benchmark models

Model	Main idea
LS	Linear no-arbitrage model
EN	Elastic-net regularized linear model
FFN	Neural-network return prediction
GAN	No-arbitrage + adversarial moments + LSTM states

Model Comparison Metrics

Metric	Construction	Role in model comparison
SDF Sharpe ratio	$SR(F) = \frac{E[F]}{\sqrt{\text{Var}(F)}}$	Measures the investment performance of the estimated SDF factor.
Explained variation	$EV = 1 - \frac{\sum_i E[\epsilon_i^2]}{\sum_i E[(R_i^e)^2]}$	Measures how much of individual stock return variation is captured by the SDF exposure.
Cross-sectional R^2	$XS-R^2 = 1 - \frac{\frac{1}{N} \sum_i E[\epsilon_i]^2}{\frac{1}{N} \sum_i E[R_i^e]^2}$	Whether the model explains the cross-section of expected returns, which is the central empirical target in asset pricing.

- **Sharpe ratio** evaluates the quality of the estimated SDF portfolio.
- **Explained variation** checks whether the SDF captures systematic return movements.
- **Cross-sectional R^2** checks whether average returns are priced with small errors.

Together, they separate investment efficiency, return decomposition, and pricing performance.

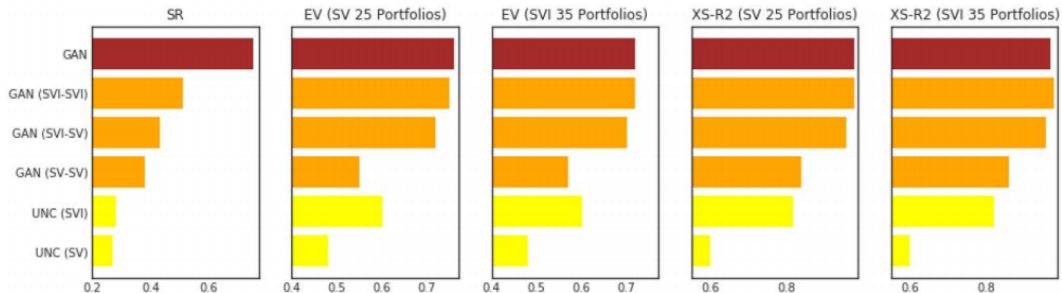
H1: No-Arbitrage Objective Improves Asset Pricing

Model	SR			EV			XS- R^2		
	Train	Valid	Test	Train	Valid	Test	Train	Valid	Test
LS	1.80	0.58	0.42	0.09	0.03	0.03	0.15	0.00	0.14
EN	1.37	1.15	0.50	0.12	0.05	0.04	0.17	0.02	0.19
FFN	0.45	0.42	0.44	0.11	0.04	0.04	0.14	-0.00	0.15
GAN	2.68	1.43	0.75	0.20	0.09	0.08	0.12	0.01	0.23

The evidence supports H1.

GAN outperforms the prediction-based FFN and linear no-arbitrage benchmarks in out-of-sample Sharpe ratio, explained variation, and cross-sectional pricing performance.

H2: Adversarial Test Assets Improve SDF Identification

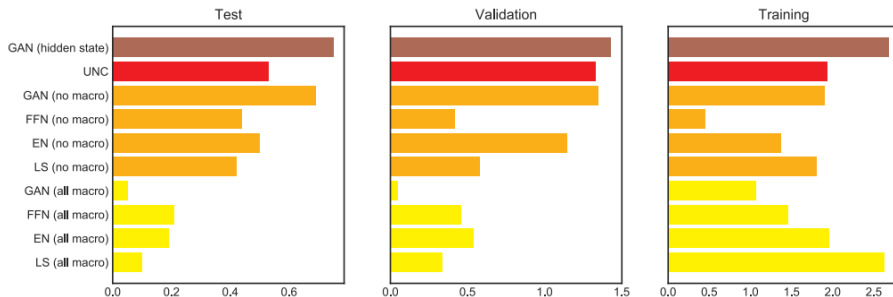


- The SDF can only learn pricing information contained in the test assets.
- Pre-specified or unconditional test assets may miss important pricing errors.
- The adversary constructs assets and states that are hardest to price.

The evidence supports H2.

GAN test assets reveal pricing info that fixed test assets may hide, improving SDF identification.

H3-A: Dynamic Macro States Matter

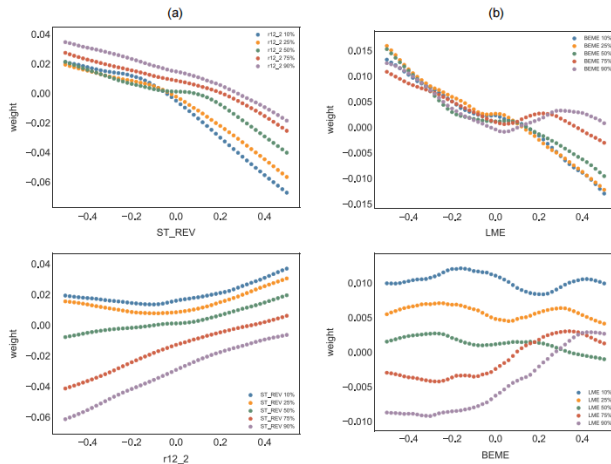


- Directly feeding all macro variables leads to weak out-of-sample performance.
- LSTM hidden states preserve dynamic information in macro time series.
- Hidden states are closely related to recessions and crisis periods.

The evidence supports H3a (macro-state)

Macro-info matters when model extracts dynamic states rather than raw increments.

H3-B: Nonlinear Characteristic Interactions Matter



- (a). Interaction between ST_REV and r12_2; (b). between Size and BEME.
- **Marginal effect of one characteristic on SDF weight depends on level of another characteristic.**

H3-B: interactions improve pricing of anomaly portfolios

Characteristic	Explained Variation			Cross-Sectional R^2		
	EN	FFN	GAN	EN	FFN	GAN
ST_REV	0.43	0.58	0.70	0.45	0.79	0.94
SUV	0.42	0.75	0.83	0.64	0.97	0.99
r12_2	0.26	0.27	0.54	0.66	0.71	0.93
LME	0.83	0.78	0.86	0.96	0.95	0.97
BEME	0.70	0.75	0.82	0.97	0.94	0.98
IdioVol	0.43	0.27	0.66	0.79	0.72	0.97
MktBeta	0.44	0.44	0.64	0.81	0.85	0.97
NI	0.51	0.60	0.75	0.88	0.96	0.99
ATO	0.58	0.70	0.77	0.96	0.98	0.99

The evidence supports H3b(nonlinear-interaction).

GAN dominates EN and FFN for representative anomaly portfolios.

Nonlinear chara interactions contain pricing info beyond linear or additive effects.

Replication: Table 1

原文	SR			EV			XS- R^2		
	Train	Valid	Test	Train	Valid	Test	Train	Valid	Test
LS	1.80	0.58	0.42	0.09	0.03	0.03	0.15	0.00	0.14
EN	1.37	1.15	0.50	0.12	0.05	0.04	0.17	0.02	0.19
FFN	0.45	0.42	0.44	0.11	0.04	0.04	0.14	-0.00	0.15
GAN	2.68	1.43	0.75	0.20	0.09	0.08	0.12	0.01	0.23

复现	SR			EV			XS- R^2		
	Train	Valid	Test	Train	Valid	Test	Train	Valid	Test
LS	1.47	0.73	0.39	0.09	0.04	0.03	0.16	0.01	0.15
EN	1.22	0.97	0.41	0.21	0.11	0.07	0.18	0.03	0.22
FFN	0.35	0.29	0.37	0.10	0.04	0.04	0.16	0.02	0.18
GAN	2.49	1.36	0.70	0.17	0.08	0.09	0.14	0.02	0.21

Replication: Figure 6

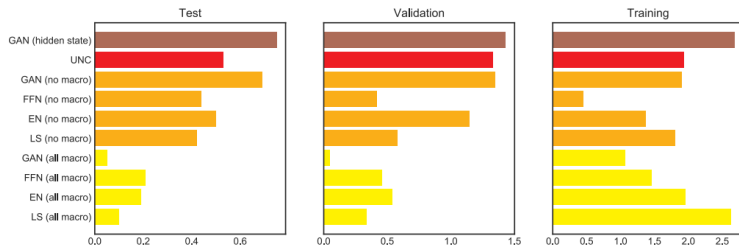
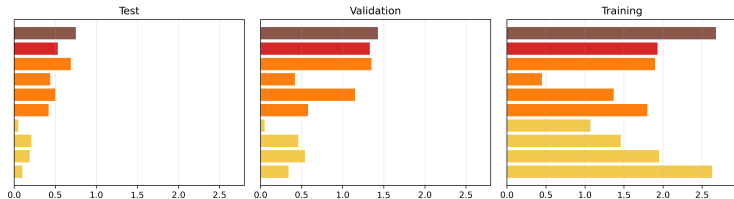
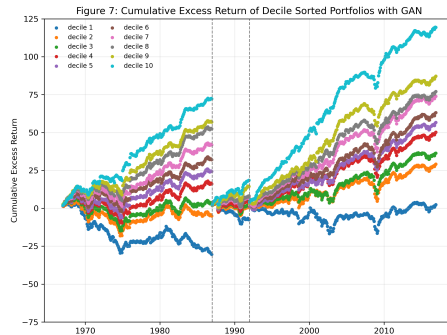
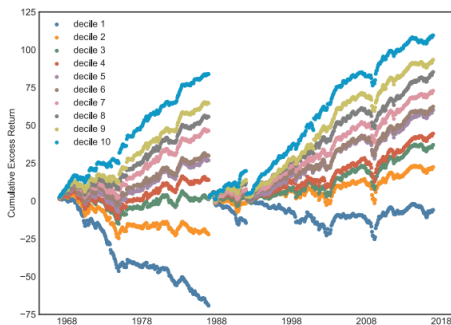


Figure 6: Performance of Models with Different Macroeconomic Variables



Replication: Figure 7

Figure 7. (Color online) Cumulative Excess Return of Decile-Sorted Portfolios with GAN



拓展

本文已经证明：在无套利约束下，深度学习可以估计条件 SDF。

是否可以拓展检验：**估计出的 SDF 是否可交易、可解释。**

1. 可实施的 SDF

- 本文主要评价 gross pricing performance。
- 可将交易成本、换手率、卖空约束和容量约束直接纳入估计目标。
- 核心问题：高 Sharpe SDF 在真实交易摩擦下是否仍然有效？

2. 可解释的经济状态

- 本文用 LSTM 提取宏观 hidden states。
- 后续可识别这些状态是否对应流动性、通胀、信用风险或政策周期。
- 核心问题：深度学习 SDF 捕捉的是哪类经济风险？

在交易摩擦、可解释经济状态下，无套利深度学习框架能否估计出更可实施、更可解释的 SDF ？

Appendix: Alternative Models as the Comparison Framework

Model	Flexible nonlinear	No-arbitrage	Main role
FFN forecasting	✓	×	Tests if deep-learning return prediction is sufficient.
Linear no-arbitrage	×	✓	Tests whether a linear asset-pricing model is sufficient.
GAN	✓	✓	Nonlinear deep learning + no-arbitrage restrictions.

GAN vs. FFN forecasting

- Both allow flexible nonlinear functions.
- FFN minimizes return prediction errors.
- GAN minimizes pricing errors.

This comparison tests the value of no-arbitrage.

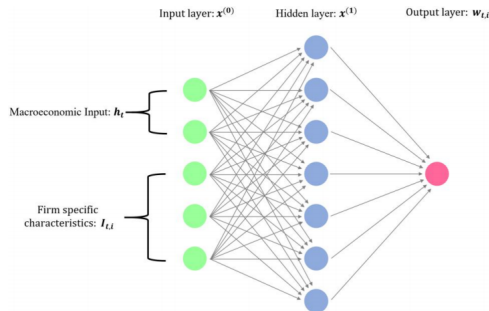
GAN vs. Linear

- Both impose no-arbitrage restrictions.
- Linear-SDF restricts to a linear function.
- GAN allows nonlinearities, interactions.

This comparison tests the value of nonlinear learning.

Appendix: FFN and LSTM

Feedforward network (FFN)



- Maps inputs into outputs through hidden layers:

$$x^{(1)} = \sigma(W^{(0)\top} x^{(0)} + w_0^{(0)}), \quad y = W^{(1)\top} x^{(1)} + w_0^{(1)}.$$
- Nonlinear activations allow the model to capture nonlinearities and interactions.
- FFNs estimate ω , g , μ , and β .

Why LSTM is needed

- Macro variables are high-dimensional and often non-stationary.
- Using only the latest ones loses dynamic info.
- Asset pricing states may depend on long-run macro patterns, such as business cycles.

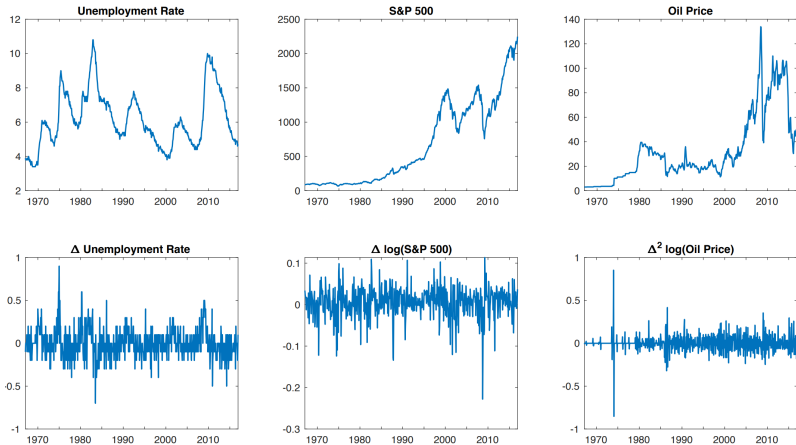
LSTM

- $(x_0, x_1, \dots, x_t) \longrightarrow h_t = h^{LSTM}(x_0, \dots, x_t)$
- Input: a sequence of transformed macro variables.
 - Output: several hidden macro states h_t .
 - Summarize short- and long-run macro dynamics.

How they enter the GAN-SDF model

- Features $I_{t,i} + \text{macro states } h_t \longrightarrow \omega(I_{t,i}, h_t)$
- FFN captures the X-S nonlinear SDF structure.
 - LSTM extracts the time-varying economic state.
 - Allow SDF to vary across firms and over time.

Appendix: Macro variables



- FRED-MD 数据库中的 124 个宏观预测变量；
- 46 个公司特征的横截面中位数时间序列；
- Welch and Goyal 提出的 8 个股权溢价预测变量。
- 对这些宏观时间序列进行标准平稳化变换，使其适合进入时间序列模型

Appendix: Model Hyperparameters

Notation	Hyperparameter	Candidates	Optimal
HL	Number of layers in SDF network	2, 3, or 4	2
HU	Number of hidden units in SDF network	64	64
SMV	Number of hidden states in SDF network	4 or 8	4
CSMV	Hidden states in conditional network	16 or 32	32
CHL	Layers in conditional network	0 or 1	0
CHU	Hidden units in conditional network	4, 8, 16, or 32	8
LR	Initial learning rate	0.001, 0.0005, 0.0002, 0.0001	0.001
DR	Dropout keep probability	0.95	0.95
D	Adversarial instruments	–	8

The benchmark GAN uses a two-layer SDF network with 64 hidden units per layer and four macro hidden states.

The adversarial network uses 32 macro hidden states and directly maps firm characteristics and macro states into eight conditional instruments.